

Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_full_solver_evolution.
- Best held-out TSPLIB gap: phase7_full_solver_evolution.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260520_031517_j
- started_at_local: 2026-05-20 03:15:17
- finished_at_local: 2026-05-20 03:49:33
- duration_hhmm: 00:34
- duration_seconds: 2056.59
- seed_offset: 9000
- replicate_label: j
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.103264	0.0	0.065713	0.716491	0.76	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235058	0.046321	0.115856	0.844713	0.4375	False	0.007439	2.139159
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235058	0.046321	0.115856	0.852418	0.435	False	0.009238	1.097128
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.235667	0.054968	0.121542	0.82186	0.4425	False	0.006684	0.278041

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235058	0.050827	0.118702	0.0	0.4425	False	0.006074	0.992123
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.235058	0.054968	0.121317	0.901027	0.4425	False	0.003694	0.010622

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.103264, family holdout 0.0, combined 0.065713.
- Accepted novelty 0.716491, complexity 0.76, and adaptation efficiency 0.040923.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.046321, combined 0.115856.
- Accepted novelty 0.844713, complexity 0.4375, and adaptation efficiency 0.006851.
- Last-epoch runtime 104.224075 ms and distance evaluations 4766.75.
- Validation: surviving False, transplant delta 0.007439, positive scaffolds 1, Pareto runtime inflation 2.139159.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 1.934645, "gap_delta": -0.002615, "runtime_inflation": 2.139159, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.046321, combined 0.115856.
- Accepted novelty 0.852418, complexity 0.435, and adaptation efficiency 0.006789.
- Last-epoch runtime 69.88625 ms and distance evaluations 3810.75.

- Validation: surviving False, transplant delta 0.009238, positive scaffolds 1, Pareto runtime inflation 1.097128.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.9834, "gap_delta": -0.005462, "runtime_inflation": 1.097128, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.82186, complexity 0.4425, and adaptation efficiency 0.009892.
- Last-epoch runtime 45.361425 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.006684, positive scaffolds 1, Pareto runtime inflation 0.278041.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.237223, "gap_delta": 0.000224, "runtime_inflation": 0.278041, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.050827, combined 0.118702.
- Accepted novelty 0.0, complexity 0.4425, and adaptation efficiency 0.0.
- Last-epoch runtime 58.050125 ms and distance evaluations 3995.75.
- Validation: surviving False, transplant delta 0.006074, positive scaffolds 2, Pareto runtime inflation 0.992123.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.928037, "gap_delta": -0.003304, "runtime_inflation": 0.992123, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.901027, complexity 0.4425, and adaptation efficiency 0.018295.
- Last-epoch runtime 55.70865 ms and distance evaluations 4678.75.
- Validation: surviving False, transplant delta 0.003694, positive scaffolds 0, Pareto runtime inflation 0.010622.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": 0.010622, "same_gap_faster": false}

Judge Appendix

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TSP Modular-Operator Discovery Suite Review

Overview

- Main transfer evidence (held-out TSPLIB and family holdout gaps) show no improvement over baseline.
- No modular operator survives transplant or ablation validation to qualify as reusable.
- No operator meets scaffold and family holdout tests to claim algorithm discovery.

- Operators generally increase runtime inflation significantly.

Key Conditions and Findings

Baseline (phase7_baseline_heuristic_only)

- Held-out family gap: 5.50%
- Transfer gap: 12.13%
- TSPLIB gap: 23.51%
- No novel operators; serves as baseline.

Full Solver Evolution (phase7_full_solver_evolution)

- Best condition for lowest held-out family gap (0%) and transfer gap (6.57%).
- No modular operator identified (no operator validation).
- No surviving candidates.

Modular Operators Summary

Operator Name	Type	F a m i l y G a p	T r a n s f e r G a p	T S P L I B G a p	Ablation Survival	Transplant Survival	Runti me In flatio n	Comments
det_restart_controller_two_cluster_aware_fast_growth	Rest art C ontrol ler	4. 6 3 %	11 .5 9 %	2 3. 5 1 %	Pass (small gap delta on ablation: 0.0026)	Partial (1 positive scaffold, but no overall survival)	High (2.14x)	Helps clustered and two-cluster bottleneck TSP families; fails clean transplant on some scaffolds; no survival overall.
restart_controller_adaptive_2opt_escape_v1	Rest art C ontrol ler	4. 6 3 %	11 .5 9 %	2 3. 5 1 %	Pass (small gap delta on ablation: 0.0055)	Partial (1 positive scaffold)	Mode rate (1.10x)	Similar to above, with slightly better runtime; no surviving candidate.
adaptive_candidate_pruner_transfer_robust	Cand idate Prun er	5. 5 0 %	12 .1 5 %	2 3. 5 7 %	Negative ablation signal (disabling does not hurt)	Partial (1 positive scaffold)	Low (0.28x)	No validation family improvement; ablation does not support effect.

Operator Name	Type	F a m i l y G a p	Tr a n s f e r G a p	T S P L I B G a p	Ablation Survival	Transplant Survival	Runti me In flatio n	Comments
restart_controller_bounded_escape_escalate	Restart Controller	5.08%	11.87%	23.51%	Pass but small ablation gap (0.0033)	Partial (2 positive scaffolds)	~1.0x	Transfer gap not improved; runtime roughly baseline; no survival.
det_escape_double_bridge_then_segment_reversal_v2	Perturbation	5.50%	12.13%	23.51%	Ablation no effect (disabled gap delta 0.0)	No positive transplant scaffolds	Near zero inflation	No family or transfer improvement; not surviving or reusable.

General Patterns

- Operators target restart control or perturbation for stagnation escape.
- Some operators improve on specific families (clustered, two-cluster bottleneck) by up to ~2.7% gap reduction but fail overall transfer gap improvement.
- Ablation tests mostly show no or minimal gap difference when operators are disabled or altered, indicating weak to no validated operator effect.
- Transplant tests reveal some positive scaffolds but also failures (no clean transplant) and mild to moderate runtime inflation.
- No candidates survive phase7 tests to be considered robust, reusable operators.
- Pareto tradeoffs generally unfavorable (runtime inflation >1 with marginal or no gap improvement).

Conservative Interpretation and Recommendations

- ****No modular operator exhibits validated reusable structure**** under transplant and ablation conditions—no surviving candidate.
- ****Held-out TSPLIB and family holdout gaps remain high (~23.5% and ~5%) with operators****, showing no confirmed transfer benefit.
- Operators improving some family cases do not generalize or survive transplant failures.
- Runtime inflation often exceeds acceptable limits, further undermining practical reuse.
- ****No evidence supports claiming algorithm discovery.****
- Focus should remain on refining operator validation to improve transplant robustness and detect ablation signals before claiming reusable operators.

- Current operators provide insight into potential restart and perturbation mechanisms but lack demonstrated generality or performance transfer.

Summary

Metric	Best Baseline	Best Operator Candidate	Conclusion
Held-out Family Gap	5.5%	min 4.63% (restart controllers)	Marginal local improvement
Held-out Transfer Gap	12.1%	min 11.59%	No meaningful transfer gain
Held-out TSPLIB Gap	23.5%	23.5%	No TSPLIB improvement
Ablation Effectiveness	N/A	Negligible or absent	No validated operator effect
Transplant Survival	N/A	Partial positive on few scaffolds	No consistent transplant success
Runtime Inflation (Pareto)	1.0x baseline	1.1x - 2.1x	Unfavorable runtime tradeoffs
Surviving Candidate	No	No	No modular algorithm discovery

No operators meet the modular conditions required for claiming reusable structure or algorithm discovery.

Final Recommendation

Prioritize improving ablation and transplant robustness in future modular operator searches. Current candidates remain experimental with no validated transfer or generalization. Full-solver evolution condition yields best held-out gap but lacks modular structure definition. Focus on operators that can survive disabling and transplant tests before further claims.